

Galaxy Classification Case Study Rubric

DS 4002 – Javier Rasero

Due: May 6, 2026

Submission format: Upload link to GitHub repository on UVA Canvas

Access the case study repository here: https://github.com/diegoamcg/DS4002_CS3.git

Why am I doing this?

This case study allows you to apply your data science and machine learning knowledge to a real-world problem in astrophysics: classifying galaxies based on image data. You will explore how a computer can be used to automate scientific classification tasks that would otherwise require significant human effort. Through this assignment, you will gain experience working with image datasets, training Convolutional Neural Network (CNN) models, and interpreting model behavior.

What am I going to do?

The GitHub repository for this case study contains a link to download a dataset of labeled galaxy images and supporting materials. You will work with the Galaxy10 DECaLS dataset, which includes over 17,000 images across 10 galaxy classes.

You will begin by exploring the dataset and understanding the different galaxy classifications. Then, using Python, you will preprocess the image data and split it into training, validation, and test sets. Next, you will build and train a CNN model, whether that be pretrained or your own model. After training your model, you will evaluate its performance using metrics such as accuracy, precision, recall, and F1-score.

To better understand model performance, you will generate a confusion matrix and analyze how predictions vary across galaxy classes. You will also use Grad-CAM (Gradient-weighted Class Activation Mapping) to visualize which regions of an image contribute most to the model's predictions, as well as ensure your model is looking in the right place!

Finally, you will interpret your results and assess whether the model successfully classifies galaxies and what limitations it may have.

Your final deliverables should include:

- A well-documented GitHub repository containing all materials
- A Jupyter Notebook with:
 - Data preprocessing
 - Model training
 - Evaluation metrics
 - Visualizations (confusion matrix, Grad-CAM examples)
- A short (~1-2 page) written analysis about your model and its results

Tips for success:

- Focus on understanding the data before modeling
- Start with a simple model before trying more complex approaches
- Pay attention to class imbalance, as some galaxy types appear less frequently
- Carefully interpret your results, accuracy alone does not tell everything

- Talk with others! Share your progress with friends and instructors and ask for help so as to not get stuck.

How will I know I have succeeded?

You will meet expectations on this case study when you successfully follow and complete the criteria in the rubric below

Formatting	• A link to your GitHub Repository with the following: ○ OUTPUTS ■ Folder containing any images/outputs generated throughout the project, including EDA, Confusion Matrix, Grad-CAM, etc. ○ SCRIPTS ■ Folder containing all scripts used for analysis, including EDA and model creation ○ README.md ○ LICENSE.md ■ This file explains to a visitor the terms under which they may use and cite your repository. Recommended: MIT License
SCRIPTS	• Well documented Jupyter Notebook file(s) that contain code used to execute EDA and the model of your choosing/creation. See starter code for example. In the code, you must include ○ Numerous comments explaining what each command/function is doing ○ Descriptive variable names (i.e. no “filamenewupdated”) to avoid any possible confusion
README.md	• You should split this up into 3 sections, starting with: ○ Section 1: A brief summary of the case study and what types of software used, any packages used, and the platform you ran this on (i.e. Windows, Linux, Mac) ○ Section 2: Outline of documentation showing what each folder holds and listing the files in those folders ○ Section 3: Instructions on reproducing your results. Write out the steps it takes to reproduce your results. Write this out in an easy to read, step by step format that anyone new to machine learning could understand.
Analysis	• In a short 1-2 page writeup, interpret your results, whatever they may be! Make sure to include: ○ 1 paragraph about the background to the purpose of this analysis. ○ 1 paragraph about the data and anything interesting you found during EDA (with at least 1 image). ○ 2 paragraphs about your results. Analyze any challenges encountered and how you went about dealing with them, results from the model, possible improvements, and the impact these results could have in the greater context of galaxy research (include 1 image like a confusion matrix or Grad-CAM image). ○ At the end, include all references used over the course of this study.